

CHARMM-GUI

Effective Simulation Input Generator and More

CHARMM is a versatile program for atomic-level simulation of many-particle systems, particularly macromolecules of biological interest. - M. Karplus

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Membrane Builder

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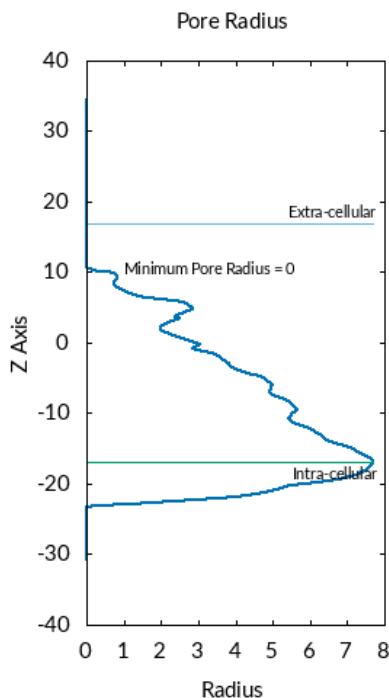
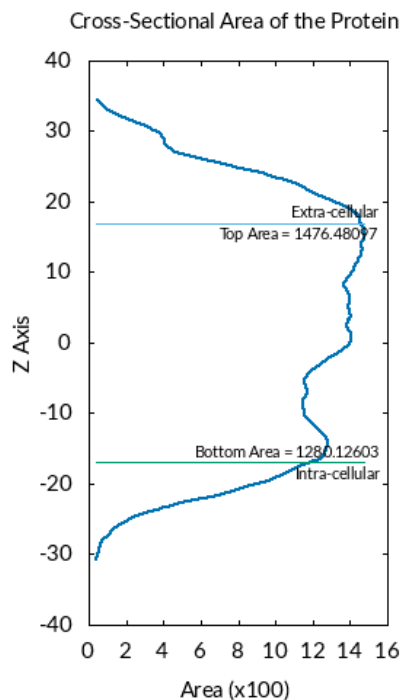
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PDB Info STEP 1 STEP 2 STEP 3 STEP 4 STEP 5 STEP 6 JOB ID: 3757172590

CHARMM PDB: [step1_pdbreader.pdb](#) ([view structure](#))
 Orientation Input: [step2_orient.inp](#)
 Orientation Output: [step2_orient.out](#)
 Oriented PDB: [step2_orient.pdb](#) ([view structure](#)) (please view this structure before you move to the next step)
 Area Calculation: [step2_area.str](#) Calculate cross sectional area of the protein
[step2_area.plo](#) Computed cross sectional area along Z axis
[step2_protein_area.str](#) Top/Bottom area of the protein
 Pore Water: [step2.1_pore.inp](#)
[step2.1_pore.out](#)
[step2.1_pore.pdb](#) ([view structure](#))
[step2.1_porewat.pdb](#) ([view structure](#))
[step2.1_porewat.crd](#)

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Calculated Cross Sectional Area:



Protein Top Area: 1476.48097
 Protein Bot Area: 1280.12603
 X Extent: -25 to 24
 Y Extent: -21 to 22
 Z Extent: -32 to 34
 Pore Minimum Radius: 0

Protein Projection onto XY:

Projection of protein portions above and below the membrane surface onto the membrane surface can be performed only to exclude pseudo lipid spheres underneath the protein portion during the lipid sphere packing.

☐ Turn on protein projection onto XY of upper leaflet

☐ Turn on protein projection onto XY of lower leaflet

This option is not recommended to use unless it is absolutely necessary

System Size Determination Options:

☐ Homogeneous Lipid "Homogeneous Lipid" option is no longer supported. You can use "Heterogeneous Lipid" option even for homogeneous lipid bilayer building.

Heterogeneous Lipid

1. Box Type: Rectangular (Currently, only CHARMM, NAMD, and GROMACS support the hexagonal box)
2. Length of Z based on:

Water thickness 15 (Minimum water height on top and bottom of the system)

Hydration number 50 (Number of water molecules per one lipid molecule)
3. Length of XY based on:

Ratios of lipid components

Numbers of lipid components

XY Dimension Ratio: 1
(The system size along the X and Y must be the same)

Show the system info click this once you fill the following table:

Lipid Type	Charge [e]	Tail Info. [sn1/sn2]	Images	# of Lipid on Upperleaflet	# of Lipid on Lowerleaflet	Surface Area
Sterols						
PA (phosphatidic acid) Lipids						
PC (phosphatidylcholine) Lipids						
DDPC	0	10:0 / 10:0	[Image]	0	0	64.1
DCPC	0	11:0 / 11:0	[Image]	0	0	64.1
DLPC	0	12:0 / 12:0	[Image]	0	0	64.1
DMPC	0	14:0 / 14:0	[Image]	0	0	64.1
DPPC	0	16:0 / 16:0	[Image]	0	0	63.0
DSPC	0	18:0 / 18:0	[Image]	0	0	65.6
PSPC	0	16:0 / 18:0	[Image]	0	0	68.3
PYPC	0	16:0 / 16:1	[Image]	0	0	64.1
POPC	0	16:0 / 18:1	[Image]	107	110	68.3
PLPC	0	16:0 / 18:2	[Image]	0	0	65.2
PILPC	0	16:0 / 18:2	[Image]	0	0	60.0
PEPC	0	16:0 / 22:1	[Image]	0	0	66.5
SOPC	0	18:0 / 18:1	[Image]	0	0	66.2
SLPC	0	18:0 / 18:2	[Image]	0	0	65.2
DRPC	0	14:1 / 14:1	[Image]	0	0	67.0
DYPC	0	16:1 / 16:1	[Image]	0	0	67.0
YOPC	0	16:1 / 18:1	[Image]	0	0	67.0
DOPC	0	18:1 / 18:1	[Image]	0	0	69.7
OEPC	0	18:1 / 22:1	[Image]	0	0	66.5
DUPC	0	18:2 / 18:2	[Image]	0	0	70.0
DGPC	0	20:1 / 20:1	[Image]	0	0	66.7
DEPC	0	22:1 / 22:1	[Image]	0	0	66.5
DNPC	0	24:1 / 24:1	[Image]	0	0	62.3
DLIPC	0	18:2 / 18:2	[Image]	0	0	65.2

- PE (phosphatidylethanolamine) Lipids
- PG (phosphatidylglycerol) Lipids
- PS (phosphatidylserine) Lipids
- PP (pyrophosphate) Lipids
- PI (phosphatidylinositol) Lipids
- CL (cardiolipin) Lipids
- DAG (diacylglycerol) Lipids ☐ Optimized charges for DAG
- TAG (triacylglycerol) Lipids ☐ Optimized charges for TAG
- TAP (trimethylammonium propane) & DAP/DMA (dimethylamino) Lipids
- PUFA (polyunsaturated fatty acid) Esterified Lipids
- SM (sphingo) and Ceramide Lipids
- Ether/Plasmalogen Lipids
- Bacterial Lipids
- Endosomal Lipids
- Ubiquinone/Ubiquinol & Archaeal Lipids
- Oxidized Lipids ?

Calculated XY System Size:		
	Upperleaflet	Lowerleaflet
Protein Area	1476.48097	1280.12603
Lipid Area	7548.1	7753
# of Lipids	115	118
Total Area	9024.58097	9033.12603
Protein X Extent	24.29	
Protein Y Extent	21.65	
Average Area	9028.85	
A	95.02	
B	95.02	

▼ Fatty Acids

LAU	-1	12:0	[Image]	0	0	30.0
MYR	-1	14:0	[Image]	0	0	30.0
PAL	-1	16:0	[Image]	0	0	30.0
STE	-1	18:0	[Image]	0	0	30.0
ARA	-1	20:0	[Image]	0	0	30.0
BEH	-1	22:0	[Image]	0	0	30.0
TRI	-1	23:0	[Image]	0	0	30.0
LIGN	-1	24:0	[Image]	0	0	30.0
MYRO	-1	14:1 (Δ^9)	[Image]	0	0	30.0
PALO	-1	16:1 (Δ^9)	[Image]	0	0	30.0
HTA	-1	16:3 ($\Delta^{7,10,13}$)	[Image]	0	0	30.0
OLE	-1	18:1 (Δ^9)	[Image]	0	0	30.0
LIN	-1	18:2 ($\Delta^{9,12}$)	[Image]	0	0	30.0
ALIN	-1	18:3 ($\Delta^{9,12,15}$)	[Image]	0	0	30.0
SDA	-1	18:4 ($\Delta^{6,9,12,15}$)	[Image]	0	0	30.0
GLA	-1	18:3 ($\Delta^{6,9,12}$)	[Image]	0	0	30.0
EICO	-1	20:1 (Δ^{11})	[Image]	0	0	30.0
EDA	-1	20:2 ($\Delta^{11,14}$)	[Image]	0	0	30.0
MEA	-1	20:3 ($\Delta^{5,8,11}$)	[Image]	0	0	30.0
DGLA	-1	20:3 ($\Delta^{8,11,14}$)	[Image]	0	0	30.0
ETE	-1	20:3 ($\Delta^{11,14,17}$)	[Image]	0	0	30.0
ETA	-1	20:4 ($\Delta^{8,11,14,17}$)	[Image]	0	0	30.0
EPA	-1	20:5 ($\Delta^{5,8,11,14,17}$)	[Image]	0	0	30.0
ARAN	-1	20:4 ($\Delta^{5,8,11,14}$)	[Image]	8	8	30.0
HPA	-1	21:5 ($\Delta^{6,9,12,15,18}$)	[Image]	0	0	30.0
ERU	-1	22:1 (Δ^{13})	[Image]	0	0	30.0
DDA	-1	22:2 ($\Delta^{13,16}$)	[Image]	0	0	30.0
ADR	-1	22:4 ($\Delta^{7,10,13,16}$)	[Image]	0	0	30.0
DPT	-1	22:5 ($\Delta^{4,7,10,13,16}$)	[Image]	0	0	30.0
DPA	-1	22:5 ($\Delta^{7,10,13,16,19}$)	[Image]	0	0	30.0
DHA	-1	22:6 ($\Delta^{4,7,10,13,16,19}$)	[Image]	0	0	30.0
NER	-1	24:1 (Δ^{15})	[Image]	0	0	30.0
TTA	-1	24:4 ($\Delta^{9,12,15,18}$)	[Image]	0	0	30.0
TPT	-1	24:5 ($\Delta^{6,9,12,15,18}$)	[Image]	0	0	30.0
TPA	-1	24:5 ($\Delta^{9,12,15,18,21}$)	[Image]	0	0	30.0
THA	-1	24:6 ($\Delta^{6,9,12,15,18,21}$)	[Image]	0	0	30.0
LAUP	0	12:0	[Image]	0	0	30.0
MYRP	0	14:0	[Image]	0	0	30.0
PALP	0	16:0	[Image]	0	0	30.0
STEP	0	18:0	[Image]	0	0	30.0
ARAP	0	20:0	[Image]	0	0	30.0
BEHP	0	22:0	[Image]	0	0	30.0
TRIP	0	23:0	[Image]	0	0	30.0
LIGNP	0	24:0	[Image]	0	0	30.0
MYROP	0	14:1 (Δ^9)	[Image]	0	0	30.0
PALOP	0	16:1 (Δ^9)	[Image]	0	0	30.0
HTAP	0	16:3 ($\Delta^{7,10,13}$)	[Image]	0	0	30.0
OLEP	0	18:1 (Δ^9)	[Image]	0	0	30.0
LINP	0	18:2 ($\Delta^{9,12}$)	[Image]	0	0	30.0
ALINP	0	18:3 ($\Delta^{9,12,15}$)	[Image]	0	0	30.0
SDAP	0	18:4 ($\Delta^{6,9,12,15}$)	[Image]	0	0	30.0
GLAP	0	18:3 ($\Delta^{6,9,12}$)	[Image]	0	0	30.0
EICOP	0	20:1 (Δ^{11})	[Image]	0	0	30.0
EDAP	0	20:2 ($\Delta^{11,14}$)	[Image]	0	0	30.0
MEAP	0	20:3 ($\Delta^{5,8,11}$)	[Image]	0	0	30.0
DGLAP	0	20:3 ($\Delta^{8,11,14}$)	[Image]	0	0	30.0
ETEP	0	20:3 ($\Delta^{11,14,17}$)	[Image]	0	0	30.0
ETAP	0	20:4 ($\Delta^{8,11,14,17}$)	[Image]	0	0	30.0
EPAP	0	20:5 ($\Delta^{5,8,11,14,17}$)	[Image]	0	0	30.0
ARANP	0	20:4 ($\Delta^{5,8,11,14}$)	[Image]	0	0	30.0
HPAP	0	21:5 ($\Delta^{6,9,12,15,18}$)	[Image]	0	0	30.0
ERUP	0	22:1 (Δ^{13})	[Image]	0	0	30.0
DDAP	0	22:2 ($\Delta^{13,16}$)	[Image]	0	0	30.0
ADRP	0	22:4 ($\Delta^{7,10,13,16}$)	[Image]	0	0	30.0
DPTP	0	22:5 ($\Delta^{4,7,10,13,16}$)	[Image]	0	0	30.0
DPAP	0	22:5 ($\Delta^{7,10,13,16,19}$)	[Image]	0	0	30.0

DHAP	0	22:6 ($\Delta^{4,7,10,13,16,19}$)	[Image]	0	0	30.0
NERP	0	24:1 (Δ^{15})	[Image]	0	0	30.0
TTAP	0	24:4 ($\Delta^{9,12,15,18}$)	[Image]	0	0	30.0
TPTP	0	24:5 ($\Delta^{6,9,12,15,18}$)	[Image]	0	0	30.0
TPAP	0	24:5 ($\Delta^{9,12,15,18,21}$)	[Image]	0	0	30.0
THAP	0	24:6 ($\Delta^{6,9,12,15,18,21}$)	[Image]	0	0	30.0

► **Detergents**

► **N-acylated Amino Acids**

► **PEG (polyethylene glycol) Lipids**

► **Glycolipids**

► **LPS (lipopolysaccharides)**

Note: The surface areas listed above are only suggested values. In the case of high cholesterol concentration, 10~20% smaller surface areas will be adopted as default values, which might be more reasonable and close to equilibrium lipid areas. Please refer the manuscript or the following link to list of default surface area we are using. [?](#)

Next Step:
Determine the System Size



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